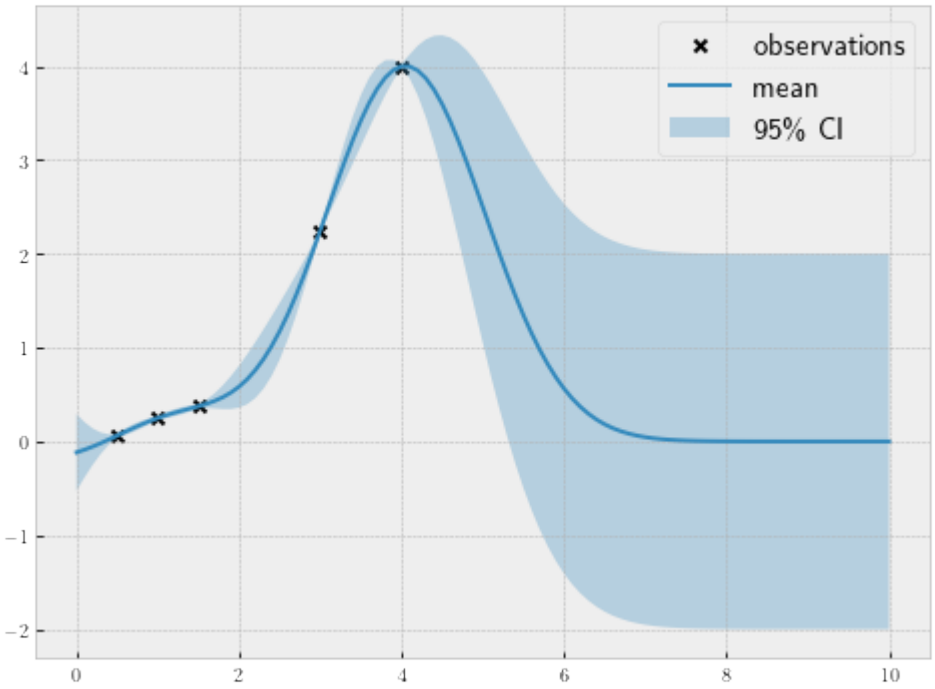


Figure 3.5 Predictions by a GP with a zero mean function. The posterior mean function interpolates the observed data points and reverts back to zero in regions that are far away from these observations.

All the nice properties about the GP that we pointed out in Section 2.4.4 are still here:

- The posterior mean function smoothly interpolates the x's that are our training data points.
- The 95% CI vanishes around these data points, denoting a well-calibrated quantification of uncertainty.

What we also notice from this plot is that once we have gone sufficiently far away from our training data points (the right side of the plot), our posterior mean function *reverts* back to the prior mean, which is zero. This is in fact an important feature of a GP: in the absence of data (in regions without observations), the prior mean function is the main driving force of the inference procedure. This makes intuitive sense, as without actual observations, the best thing that a predictive model could do is to simply appeal to the prior knowledge that is encoded in its mean function.

NOTE **At this point, we see why having well-defined prior knowledge encoded into the prior GP is so important: in the absence of data, the only thing that drives predictions is the prior GP.**

A natural question then arises: Is it possible to use a non-zero mean function to induce a different behavior for our GP in these unexplored regions, and if so, what are our options? The remaining portion of this section aims to answer this question. We will start by using a constant mean function that is not constant at zero.

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